# Nicola Dell'Avvocato

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## PROFILE

Aerospace Engineering graduate student oriented towards space missions, on-board sensing and aerospace defence. Ambitious, determined and used to leading multidisciplinary teams towards results that are measured rather than described. Seeking a career in the international space and defence industry.

## EDUCATION

**MSc in Space Engineering** Sep 2026 – expected Jul 2028
*Politecnico di Torino | Turin, Italy*

**BSc in Aerospace Engineering** Sep 2023 – Jul 2026
*Politecnico di Torino | Turin, Italy*

- Final grade: 110/110 cum Laude.

**Erasmus+ exchange semester** Feb 2026 – Jun 2026
*Universidade da Coruña | Ferrol, Spain*

**Merit-based University College — Supplementary Training Programme** Sep 2025 – Jul 2028
*Collegio Universitario di Torino Renato Einaudi | Turin, Italy*

**Scientific high school diploma — Cambridge International Curriculum** Sep 2018 – Jul 2023
*Liceo Scientifico Federico II di Svevia | Altamura, Italy*

- Final grade: 100/100.

## RESEARCH & PROJECTS

**Challenge@PoliTO by Leonardo — Simulation & Experimental Validation** (upcoming) Feb 2027 – Jun 2027
*Politecnico di Torino & Leonardo | selected for a multidisciplinary company-led challenge | Turin, Italy*

- Analysis and optimisation of electromagnetic and thermal signatures in complex physical systems.
- Development and validation of numerical models, experimental testing techniques and innovative material solutions, combining simulation, laboratory testing and data analysis in a real-world research environment.

**F450 Quadcopter with AI Mission Interface** (personal project, in progress) Oct 2026 – present
*Personal project | additive manufacturing and autonomous flight*

- Design and build of an F450-class quadcopter with additively manufactured structural parts, integrating airframe, avionics and software.
- Integration of an AI chat interface that translates natural-language input into autonomous mission configurations.

**CASSINI Hackathon — EU Space for Peace and Resilience** (in progress) Nov 2026
*European Commission / EUSPA | 12th edition*

- Developing a team solution built on EU satellite data (Copernicus Earth observation and Galileo positioning) for a peace and resilience challenge.

**Embedded FBG Sensors for SHM of CFRP Sandwich Structures** (BSc thesis) Jun 2026
*Politecnico di Torino | Turin, Italy*

- Embedded a Fibre Bragg Grating optical fibre directly between the carbon plies of a CFRP sample.
- Built an Abaqus/CAE finite element model of the panel to predict its natural frequencies and strain mode shapes.
- Ran an impact-hammer test campaign with an optical interrogator and processed the strain signals through FFT in MATLAB.

**Dynamic Characterisation of 3D-Printed Metamaterials** (Erasmus project) Feb 2026 – May 2026
*Universidade da Coruña | Ferrol, Spain | aeroelastic applications*

- Characterised additively manufactured metamaterial specimens on a dynamic mechanical analysis machine, measuring their damping response as a function of frequency.

**INTRAPRENDENTI — Smart Infrastructures** (Honour School) Sep 2024 – Jul 2025
*Politecnico di Torino | innovation and entrepreneurship programme*

- Worked in a multidisciplinary team (engineering, architecture, design) on a smart-infrastructures challenge.

## TECHNICAL SKILLS

**Simulation & CAD:** Abaqus/CAE, finite element analysis (FEA), modal analysis, SolidWorks, CAD/CAM software, 3D modelling, ESA DRAMA (space debris risk assessment)

**Programming & data:** MATLAB, Simulink, Python(libraries: basic knowledge of RocketPy), digital signal processing (DSP), FFT and spectral analysis, data analysis, LaTeX, Microsoft Office (Excel, Word, PowerPoint)

**Experimental & laboratory:** CFRP composite lay-up and manufacturing, Fibre Bragg Grating (FBG) optical sensors, optical interrogators, impact-hammer modal testing, dynamic mechanical analysis (DMA), strain measurement, structural health monitoring, test-data post-processing

**Engineering domains:** space systems, space debris, composite structures, aerodynamics, aeroelasticity, metamaterials and additive manufacturing

**Transferable:** team leadership and coordination, project management, multidisciplinary teamwork, analytical problem solving, critical thinking, interpersonal communication, community development, work in international environments

**Areas of interest:** space and military defence, dual-use and UAV technologies, optical fibre sensors, Earth observation, satellite networks and constellations, CubeSats, AI applied to aerospace, innovation and entrepreneurship

## EXPERIENCE & ACTIVITIES

**System Engineer — ICARUS PoliTo** (in progress) Oct 2026 – present
*Politecnico di Torino | student team | Turin, Italy*

- System engineering on RA (Record Aircraft), the team's solar-powered fixed-wing UAV designed for long-endurance flight on solar energy alone.
- Work across requirements, interfaces between subsystems and verification within a large multidisciplinary student team.

**Computer Laboratory Technician** Sep 2026 – present
*Politecnico di Torino | Turin, Italy | part-time*

- Technical support in the University's computer laboratories (LAIB), assisting students and staff with workstations and laboratory software.

**Front Desk Receptionist** Sep 2026 – present
*Collegio Universitario di Torino Renato Einaudi | Turin, Italy | part-time*

- Front desk shifts at the merit-based residential college: reception, assistance to residents and visitors, and access management.

**Staff** Jun 2026 – Oct 2026
*Wave by Vento (formerly Italian Tech Week) | Turin, Italy*

- Member of the organising staff of one of Europe's leading technology conferences, which brings together founders, investors and technology leaders.

**Head Lifeguard** Jun 2023 – Sep 2023
*Tenuta Giannini | Sannicandro di Bari, Italy*

- Responsible for pool safety, coordinating the lifeguard team and enforcing safety procedures with the public.

## LANGUAGES

**Italian:** native

**English:** professional working proficiency

**Spanish:** limited working proficiency

## CERTIFICATIONS & HONOURS

- **INTRAPRENDENTI** — Politecnico di Torino programme for innovative engineers, architects, planners and designers.
- **B2 DELE** (Spanish), Universidade da Coruña
- **Cambridge International Curriculum IGCSE** ($A^*$ - Chemistry, $A^*$ - Latin) · **B2 First** (English), Cambridge International Education.
- **Lifeguard Certification** — Federazione Italiana Nuoto (FIN).
- **Basic Life Support & Defibrillation (BLSD/AED)** — Croce Rossa Italiana.